

# Overview

This note summarizes the PD designs used in Lab 3 for the hummingbird simulation and records the specific formulas used to compute gains from time-domain specs. All math is LaTeX to render correctly when exported to PDF (tested with pandoc and typical VS Code PDF exporters that support MathJax/KaTeX).

## Models

Longitudinal (about  $\theta_e = 0$ ):

$$\begin{aligned} \ddot{\theta}(t) &= b_\theta \ddot{F}(t), b_\theta = \ell T m_1 \ell_1^2 + m_2 \ell_2^2 + J_1 y + J_2 y, F_e = (m_1 \ell_1 + m_2 \ell_2) g \ell T. \\ b_\theta &= \frac{\ell T}{m_1 \ell_1^2 + m_2 \ell_2^2 + J_1 y + J_2 y}, \\ F_e &= \frac{(m_1 \ell_1 + m_2 \ell_2) g}{\ell T}. \end{aligned}$$

$$b_\theta F_e = b_\theta \ddot{F}(t), = m_1 \ell_1^2 + m_2 \ell_2^2 + J_1 y + J_2 y \ell T, = \ell T (m_1 \ell_1 + m_2 \ell_2) g.$$

Lateral (successive loop closure, inner roll faster than outer yaw):

$$\begin{aligned} \ddot{\phi}(t) &= J_1 \ddot{x}(t), \ddot{\psi}(t) \approx b_\psi \ddot{\phi}(t), b_\psi \approx \ell T F_e J_{zz}, \\ J_{zz} &= m_1 \ell_1^2 + m_2 \ell_2^2 + J_2 z + J_1 z + m_3 (\ell_3 x^2 + \ell_3 y^2) + J_3 z. \\ \ddot{\phi}(t) &\approx \frac{1}{J_1} \ddot{x}(t), \ddot{\psi}(t) \approx b_\psi \ddot{\phi}(t), \\ b_\psi &\approx \frac{\ell T F_e}{J_{zz}}, \\ J_{zz} &= m_1 \ell_1^2 + m_2 \ell_2^2 + J_2 z + J_1 z + m_3 (\ell_3 x^2 + \ell_3 y^2) + J_3 z. \end{aligned}$$

$$= J_1 \ddot{x}(t), \approx b_\psi \ddot{\phi}(t), \quad b_\psi \approx J_{zz} \ell T F_e, = m_1 \ell_1^2 + m_2 \ell_2^2 + J_2 z + J_1 z + m_3 (\ell_3 x^2 + \ell_3 y^2) + J_3 z.$$

## Dirty Derivative (First-Order Filtered Difference)

For any signal  $x$ :

$$\begin{aligned} \hat{\dot{x}}[k] &= \alpha \hat{\dot{x}}[k-1] + (1-\alpha) x[k] - x[k-1] T_s, \alpha = \tau_d \tau_d + T_s. \\ \hat{\dot{x}}[k] &= \alpha \hat{\dot{x}}[k-1] + (1-\alpha) x[k] - x[k-1] T_s, \\ \alpha &= \frac{\tau_d}{\tau_d + T_s}. \end{aligned}$$

# Gain Mapping from Specs

With desired  $\zeta$  and rise time  $t_{rtr}$ :

$$\omega_n \approx 2.2 t_r. \quad \omega_n \approx \frac{2.2}{t_r}. \quad \omega_n \approx 2.2.$$

- Longitudinal (pitch):

$$k_{P\theta} = \omega_n^2 b_\theta, k_{D\theta} = 2\zeta \omega_n b_\theta. \quad k_{P_\theta} = \frac{\omega_n^2}{b_\theta}, \quad k_{D_\theta} = \frac{2\zeta \omega_n}{b_\theta}. \quad k_{P\theta} = b_\theta \omega_n^2, \quad k_{D\theta} = b_\theta 2\zeta \omega_n.$$

- Inner roll:

$$k_{P\phi} = \omega_n^2 J_{1x}, k_{D\phi} = 2\zeta \omega_n J_{1x}. \quad k_{P_\phi} = \frac{\omega_n^2}{J_{1x}}, \quad k_{D_\phi} = \frac{2\zeta \omega_n}{J_{1x}}. \quad k_{P\phi} = J_{1x} \omega_n^2, \quad k_{D\phi} = J_{1x} 2\zeta \omega_n.$$

- Outer yaw:

$$k_{P\psi} = \omega_n^2 b_\psi, k_{D\psi} = 2\zeta \omega_n b_\psi. \quad k_{P_\psi} = \frac{\omega_n^2}{b_\psi}, \quad k_{D_\psi} = \frac{2\zeta \omega_n}{b_\psi}. \quad k_{P\psi} = b_\psi \omega_n^2, \quad k_{D\psi} = b_\psi 2\zeta \omega_n.$$

## Bandwidth Separation

Choose inner-loop roll at least one order faster than yaw:

$$\omega_n \phi = M \omega_n \psi, M \in [10, 20]. \quad \omega_n \phi = M \omega_n \psi, \quad M \in [10, 20]. \quad \omega_n \phi = M \omega_n \psi, \quad M \in [10, 20].$$

## Control Laws (PD Only)

- Longitudinal PD (force):

$$\begin{aligned} e_\theta &= \theta_d - \theta, \quad \dot{e}_\theta = -\dot{\theta}, \quad \ddot{e}_\theta = \ddot{\theta}_d - \ddot{\theta}, \quad F_\sim = k_{P\theta} e_\theta + k_{D\theta} \dot{e}_\theta, \quad F = F_e + F_\sim. \\ \begin{aligned} e_\theta &= \theta_d - \theta, \\ \dot{e}_\theta &= -\dot{\theta}, \\ \ddot{e}_\theta &= \ddot{\theta}_d - \ddot{\theta}, \\ F &= k_{P_\theta} e_\theta + k_{D_\theta} \dot{e}_\theta, \\ F &= F_e + F_\sim. \end{aligned} \end{aligned}$$

$$\theta_F \sim F = \theta_d - \theta, \dot{\theta} = -\dot{\theta}$$

$$, \quad \dot{\theta} = k_P \theta e_\theta + k_D \dot{\theta} e_\theta$$

$$\theta, \quad \dot{\theta} = F_e + F_\sim.$$

- Lateral PD (successive loops):  
Outer yaw (commands roll reference):

$$e_\psi = \psi_d - \psi, \dot{e}_\psi = -\dot{\psi}, \phi_d = k_P \psi e_\psi + k_D \dot{\psi} e_\psi. e_\psi = \psi_d - \psi, \quad \widehat{\dot{e}_\psi} = -\widehat{\dot{\psi}}, \quad \phi_d = k_{\{P_\psi\}} e_\psi + k_{\{D_\psi\}} \widehat{\dot{e}_\psi}. e_\psi = \psi_d - \psi, \quad \dot{e}_\psi = -\dot{\psi}$$

$$\psi =$$

$$\dot{\psi}$$

$$, \quad \phi_d = k_P \psi e_\psi + k_D \dot{\psi} e_\psi$$

$$\psi.$$

Inner roll (commands torque):

$$e_\phi = \phi_d - \phi, \dot{e}_\phi = -\dot{\phi}, \tau = k_P \phi e_\phi + k_D \dot{\phi} e_\phi. e_\phi = \phi_d - \phi, \quad \widehat{\dot{e}_\phi} = -\widehat{\dot{\phi}}, \quad \tau = k_{\{P_\phi\}} e_\phi + k_{\{D_\phi\}} \widehat{\dot{e}_\phi}. e_\phi = \phi_d - \phi, \quad \dot{e}_\phi = -\dot{\phi}$$

$$\phi =$$

$$\dot{\phi}$$

$$, \quad \tau = k_P \phi e_\phi + k_D \dot{\phi} e_\phi$$

$$\phi.$$

## Simulation Notes

- H.7: Step  $\theta_d$  by  $\pm 5^\circ$  and verify well-damped tracking. The logged force  $F_{FF}$  should equal  $F_e F_{eFe}$  plus PD action.
- H.8: Step  $\psi_d$  (e.g.,  $30^\circ$ ). The roll response should be visibly faster; the rise-time ratio should be near MMM.

# How to Run

From the repository root (recommended to ensure imports resolve):

```
python -m _hummingbird_sim.h7_hummingbirdSim
python -m _hummingbird_sim.h8_hummingbirdSim
```

Or from the directory itself (matches current import style):

```
cd _hummingbird_sim
python h7_hummingbirdSim.py
python h8_hummingbirdSim.py
```

Python version used during development: 3.13 . If matplotlib/pyqt are missing, install from the repo root:

```
python -m pip install --user -r requirements.txt
```

## Results and Figures (placeholders)

GUI Simulation Output from H.7 & H.8

Images are expected in `../figs/` relative to this Markdown file (as you currently have in the repo).

The YAML header sets a LaTeX `\graphicspath` and a Pandoc `resource-path` so PDF export can find them even if you run Pandoc from the repo root.

### H.7 - Longitudinal PD (Pitch)

H.7 pitch tracking ( $\theta$  vs  $\theta_d$ )

H.7 force command  $F$

### H.8 - Lateral PD (Yaw/Roll)

H.8 yaw tracking ( $\psi$  vs  $\psi_d$ )

H.8 roll inner-loop ( $\phi$  vs  $\phi_d$ )

Tip: In matplotlib windows you can use the save icon, or temporarily add

```
plt.savefig('figs/h7_theta.png', dpi=200, bbox_inches='tight')
```

 near the end of the script.

# Repo Map (for this lab)

- Controllers: `ctrlLonPD.py` , `ctrlPD.py`
- Sims/plots: `h7_hummingbirdSim.py` , `h8_hummingbirdSim.py` , `dataPlotter.py`
- Dynamics/params: `hummingbirdDynamics.py` , `hummingbirdParam.py` , `signalGenerator.py`
- This report: `Lab3_Review.md` , lessons: `Lab3_Learned.txt`

## Troubleshooting

- If you see OpenGL/pyqtgraph warnings during animation, they are typically harmless on headless or constrained GPUs.
- If `_hummingbird_sim` imports fail when using `-m` , run from inside the folder with `python h7_hummingbirdSim.py` .

the 1990s, the number of people in the UK who are employed in the public sector has increased by 1.5 million, from 2.5 million in 1980 to 4 million in 1995. The public sector has become a major employer in the UK, and its growth has been a major factor in the overall growth of the economy.

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